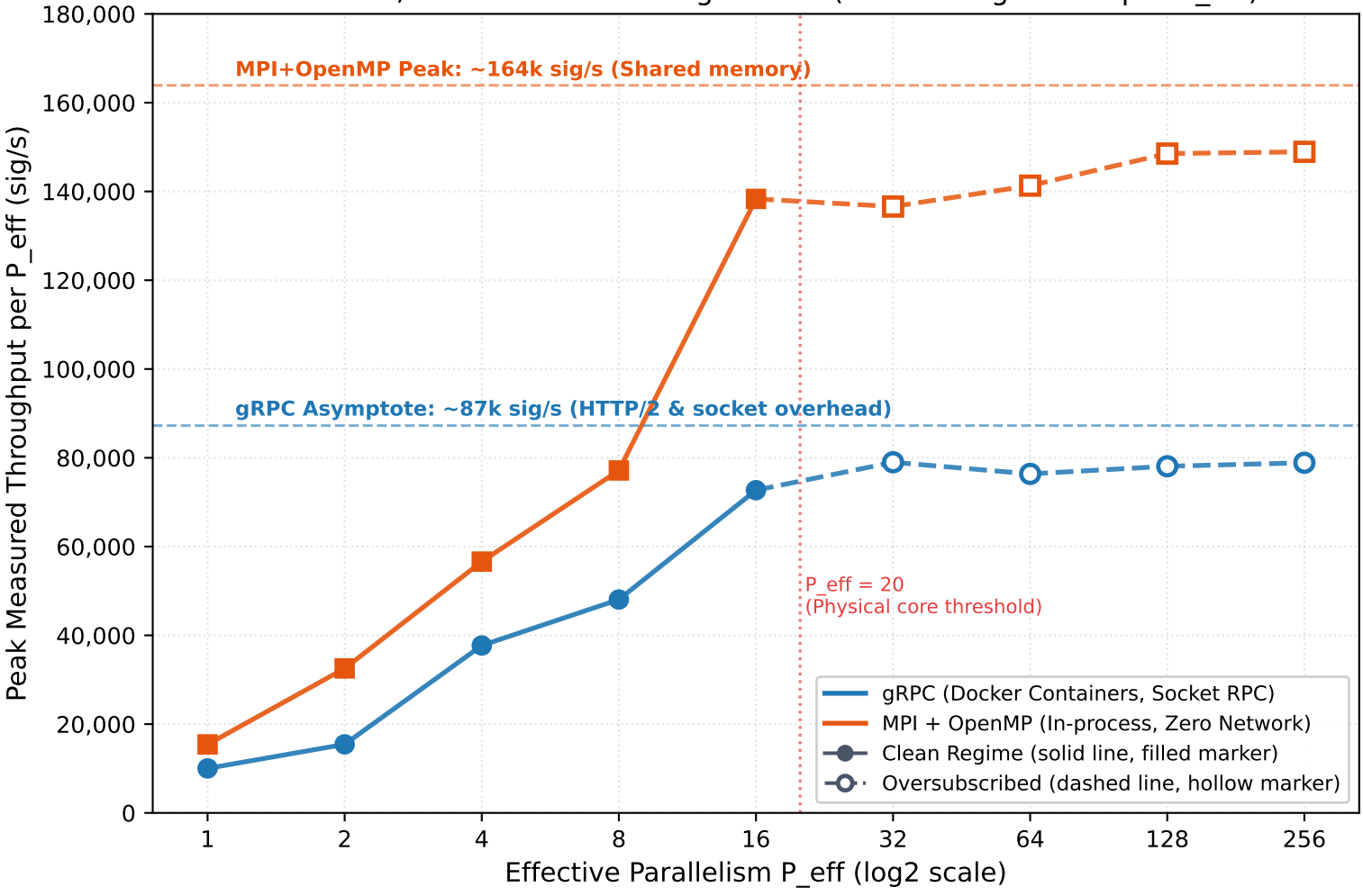
Execution Substrate Comparison: gRPC vs MPI+OpenMP Throughput
Batch = 5,000 ECDSA P-256 Signatures (Best Configuration per P_eff)
MPI+OpenMP Peak: ~164k sig/s (Shared memory)
gRPC Asymptote: ~87k sig/s (HTTP/2 & socket overhead)
P_eff = 20
(Physical core threshold)
gRPC (Docker Containers, Socket RPC)
MPI + OpenMP (In-process, Zero Network)
Clean Regime (solid line, filled marker)
Oversubscribed (dashed line, hollow marker)
Peak Measured Throughput per P_eff (sig/s)
Effective Parallelism P_eff (log2 scale)
0
20,000
40,000
60,000
80,000
100,000
120,000
140,000
160,000
180,000
1
2
4
8
16
32
64
128
256